

Wave 1 Statistical Analyses

Marta Topor & Bertil Wegmann

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Hierarchical Bayesian Models Wave 1

This document runs the models used to analyse the main outcomes for Wave 1.
The structure of the document follows the Analysis_Plans.txt file.
The variables in the datasets are explained in Meta-Data.txt.

Data Set 1 - Binary Categorical Outcomes

Crisis aware

Bayesian Logistic regression for whether participants were aware of the reproducibility crisis or not (yes/no).
The explainer variables are:

- * Whether the participant was at a university that was affiliated with the UKRN
- * Whether the participant was in their first, second, placement or third year
- * Interaction between the two variables above

```
## Inference for Stan model: anon_model.
## 4 chains, each with iter=10000; warmup=5000; thin=1;
## post-warmup draws per chain=5000, total post-warmup draws=20000.
##
##           mean se_mean   sd    2.5%    25%    50%    75%   97.5% n_eff
## mu           0.10    0.01  0.45   -0.79   -0.20    0.10    0.40    0.97  2855
## sigma        1.12    0.00  0.25    0.72    0.95    1.09    1.27    1.69  8975
## alpha[1]     -0.80    0.01  0.64   -2.09   -1.22   -0.80   -0.37    0.45  2931
## alpha[2]      0.90    0.01  0.77   -0.60    0.38    0.89    1.41    2.47  4371
## alpha[3]      0.26    0.01  0.70   -1.12   -0.19    0.26    0.73    1.63  3288
## alpha[4]      0.27    0.01  0.73   -1.17   -0.22    0.27    0.76    1.70  3818
## alpha[5]     -1.38    0.01  0.75   -2.94   -1.87   -1.36   -0.88    0.05  3886
## alpha[6]      0.86    0.01  0.82   -0.68    0.31    0.84    1.39    2.52  4967
## alpha[7]     -1.22    0.01  0.77   -2.78   -1.71   -1.20   -0.70    0.24  4335
## alpha[8]      1.88    0.01  0.84    0.33    1.30    1.84    2.41    3.62  4923
## alpha[9]     -0.10    0.01  0.58   -1.22   -0.49   -0.10    0.29    1.04  6597
## alpha[10]     2.02    0.01  0.87    0.50    1.42    1.95    2.55    3.94  8031
## alpha[11]     0.54    0.01  0.70   -0.81    0.06    0.53    1.01    1.94  6155
## alpha[12]    -1.46    0.01  0.76   -3.06   -1.94   -1.41   -0.94   -0.09  7773
## alpha[13]     0.38    0.01  0.52   -0.64    0.03    0.38    0.73    1.42  6324
## alpha[14]     0.00    0.01  0.53   -1.05   -0.36    0.00    0.35    1.03  5794
## alpha[15]    -1.26    0.01  0.69   -2.70   -1.71   -1.23   -0.79    0.01  6894
## alpha[16]     0.96    0.01  0.44    0.10    0.66    0.96    1.26    1.85  5381
## alpha[17]    -0.17    0.01  0.49   -1.14   -0.51   -0.17    0.16    0.78  6571
## alpha[18]    -1.19    0.01  0.47   -2.13   -1.49   -1.18   -0.87   -0.30  6592
## alpha[19]     0.27    0.01  0.57   -0.84   -0.12    0.27    0.66    1.40  5570
```

```

## alpha[20]    0.54    0.01 0.40   -0.25    0.26    0.54    0.81    1.32 3899
## alpha[21]    0.37    0.01 0.42   -0.44    0.09    0.38    0.65    1.20 5029
## alpha[22]    0.41    0.01 0.41   -0.38    0.13    0.41    0.69    1.22 4040
## alpha[23]    0.10    0.01 0.66   -1.18   -0.34    0.10    0.55    1.40 7780
## beta1       -0.38    0.01 0.69   -1.73   -0.84   -0.39    0.08    1.02 3114
## beta2        0.07    0.00 0.15   -0.22   -0.03    0.07    0.17    0.36 2712
## beta3        0.80    0.00 0.24    0.34    0.64    0.80    0.97    1.28 4965
## lp__        -477.30    0.05 4.10 -486.21 -479.90 -476.99 -474.40 -470.15 5786
##              Rhat
## mu              1
## sigma           1
## alpha[1]        1
## alpha[2]        1
## alpha[3]        1
## alpha[4]        1
## alpha[5]        1
## alpha[6]        1
## alpha[7]        1
## alpha[8]        1
## alpha[9]        1
## alpha[10]       1
## alpha[11]       1
## alpha[12]       1
## alpha[13]       1
## alpha[14]       1
## alpha[15]       1
## alpha[16]       1
## alpha[17]       1
## alpha[18]       1
## alpha[19]       1
## alpha[20]       1
## alpha[21]       1
## alpha[22]       1
## alpha[23]       1
## beta1           1
## beta2           1
## beta3           1
## lp__            1
##
## Samples were drawn using NUTS(diag_e) at Mon Sep 23 01:57:54 2024.
## For each parameter, n_eff is a crude measure of effective sample size,
## and Rhat is the potential scale reduction factor on split chains (at
## convergence, Rhat=1).

```

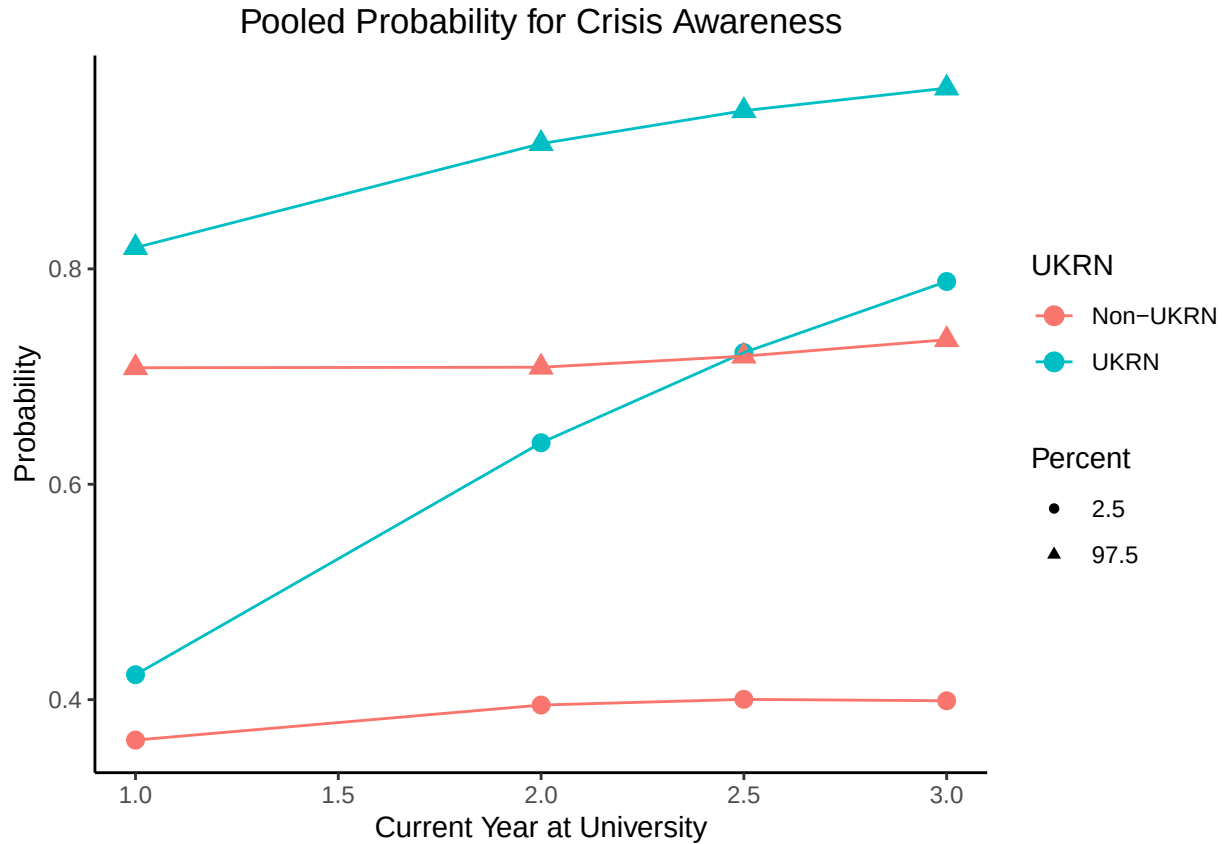
Number of participants in the analysis

Total N = 803
 UKRN N = 421
 Non-UKRN N = 382
 Year 1 N = 326
 Year 2 N = 261
 Placement Year N = 57
 Year 4 N = 159

The output shows that beta credibility intervals do not cross 0 only for the interaction term.

2.5% = 0.3369502
97.5% = 1.278575

The interpretation is that students who attend a UKRN affiliated university are more likely to be aware of the reproducibility crisis in the later years of education. For students who attend a non-UKRN affiliated university, there is no difference.



Learned about Open Science at Uni

Bayesian Logistic regression for whether participants received teaching about open science or not (yes/no).
The explainer variables are:

* Whether the participant was at a university that was affiliated with the UKRN

* Whether the participant was in their first, second, placement of third year * Interaction between the two variables above

```
## Inference for Stan model: anon_model.
## 4 chains, each with iter=10000; warmup=5000; thin=1;
## post-warmup draws per chain=5000, total post-warmup draws=20000.
##
##               mean se_mean  sd   2.5%   25%   50%   75%   97.5% n_eff
## mu           -2.08    0.01 0.48  -3.07  -2.39  -2.07  -1.75  -1.15  1829
## sigma         0.77    0.00 0.21   0.42   0.62   0.74   0.88   1.26  4974
## alpha[1]     -2.65    0.01 0.60  -3.86  -3.04  -2.64  -2.26  -1.50  2076
## alpha[2]     -1.67    0.01 0.65  -2.96  -2.10  -1.66  -1.23  -0.38  2628
## alpha[3]     -2.23    0.01 0.63  -3.50  -2.65  -2.22  -1.81  -1.01  2389
## alpha[4]     -1.61    0.01 0.65  -2.89  -2.05  -1.61  -1.17  -0.33  2570
## alpha[5]     -2.93    0.01 0.78  -4.59  -3.43  -2.89  -2.39  -1.50  3079
```

```

## alpha[6]      -1.41      0.01 0.67   -2.72   -1.86   -1.41   -0.96   -0.09  2684
## alpha[7]      -3.05      0.01 0.75   -4.61   -3.53   -3.01   -2.54   -1.68  2934
## alpha[8]      -1.09      0.01 0.62   -2.31   -1.51   -1.09   -0.67    0.13  2308
## alpha[9]      -1.60      0.01 0.65   -2.89   -2.03   -1.59   -1.16   -0.33  3639
## alpha[10]     -1.55      0.01 0.67   -2.89   -2.00   -1.54   -1.09   -0.24  3673
## alpha[11]     -2.68      0.01 0.84   -4.48   -3.21   -2.62   -2.10   -1.18  4127
## alpha[12]     -2.64      0.01 0.82   -4.42   -3.14   -2.58   -2.07   -1.18  4375
## alpha[13]     -2.00      0.01 0.64   -3.28   -2.42   -1.99   -1.57   -0.79  3566
## alpha[14]     -2.27      0.01 0.68   -3.67   -2.71   -2.25   -1.80   -1.00  3616
## alpha[15]     -2.04      0.01 0.70   -3.46   -2.49   -2.02   -1.57   -0.69  3600
## alpha[16]     -2.23      0.01 0.56   -3.37   -2.60   -2.22   -1.85   -1.18  3259
## alpha[17]     -1.66      0.01 0.58   -2.84   -2.04   -1.65   -1.27   -0.54  3627
## alpha[18]     -3.11      0.01 0.77   -4.83   -3.57   -3.05   -2.57   -1.77  4069
## alpha[19]     -1.80      0.01 0.66   -3.12   -2.24   -1.79   -1.35   -0.53  3315
## alpha[20]     -1.77      0.01 0.51   -2.79   -2.11   -1.75   -1.42   -0.78  2528
## alpha[21]     -1.52      0.01 0.51   -2.53   -1.86   -1.52   -1.17   -0.56  3010
## alpha[22]     -2.52      0.01 0.57   -3.66   -2.90   -2.50   -2.13   -1.43  2931
## alpha[23]     -1.82      0.01 0.73   -3.26   -2.30   -1.80   -1.33   -0.39  3880
## beta1          0.29      0.01 0.65   -0.99   -0.14    0.29    0.73    1.59  2360
## beta2          0.13      0.00 0.20   -0.25   -0.01    0.13    0.26    0.51  1921
## beta3          0.42      0.00 0.26   -0.07    0.25    0.42    0.59    0.93  2936
## lp__          -384.75     0.08 4.81 -395.05 -387.73 -384.45 -381.40 -376.20 4020
##               Rhat
## mu              1
## sigma           1
## alpha[1]        1
## alpha[2]        1
## alpha[3]        1
## alpha[4]        1
## alpha[5]        1
## alpha[6]        1
## alpha[7]        1
## alpha[8]        1
## alpha[9]        1
## alpha[10]       1
## alpha[11]       1
## alpha[12]       1
## alpha[13]       1
## alpha[14]       1
## alpha[15]       1
## alpha[16]       1
## alpha[17]       1
## alpha[18]       1
## alpha[19]       1
## alpha[20]       1
## alpha[21]       1
## alpha[22]       1
## alpha[23]       1
## beta1           1
## beta2           1
## beta3           1
## lp__            1
##
## Samples were drawn using NUTS(diag_e) at Mon Sep 23 02:00:44 2024.

```

```
## For each parameter, n_eff is a crude measure of effective sample size,
## and Rhat is the potential scale reduction factor on split chains (at
## convergence, Rhat=1).
```

Number of participants in the analysis

Total N = 798

UKRN N = 420

Non-UKRN N = 378

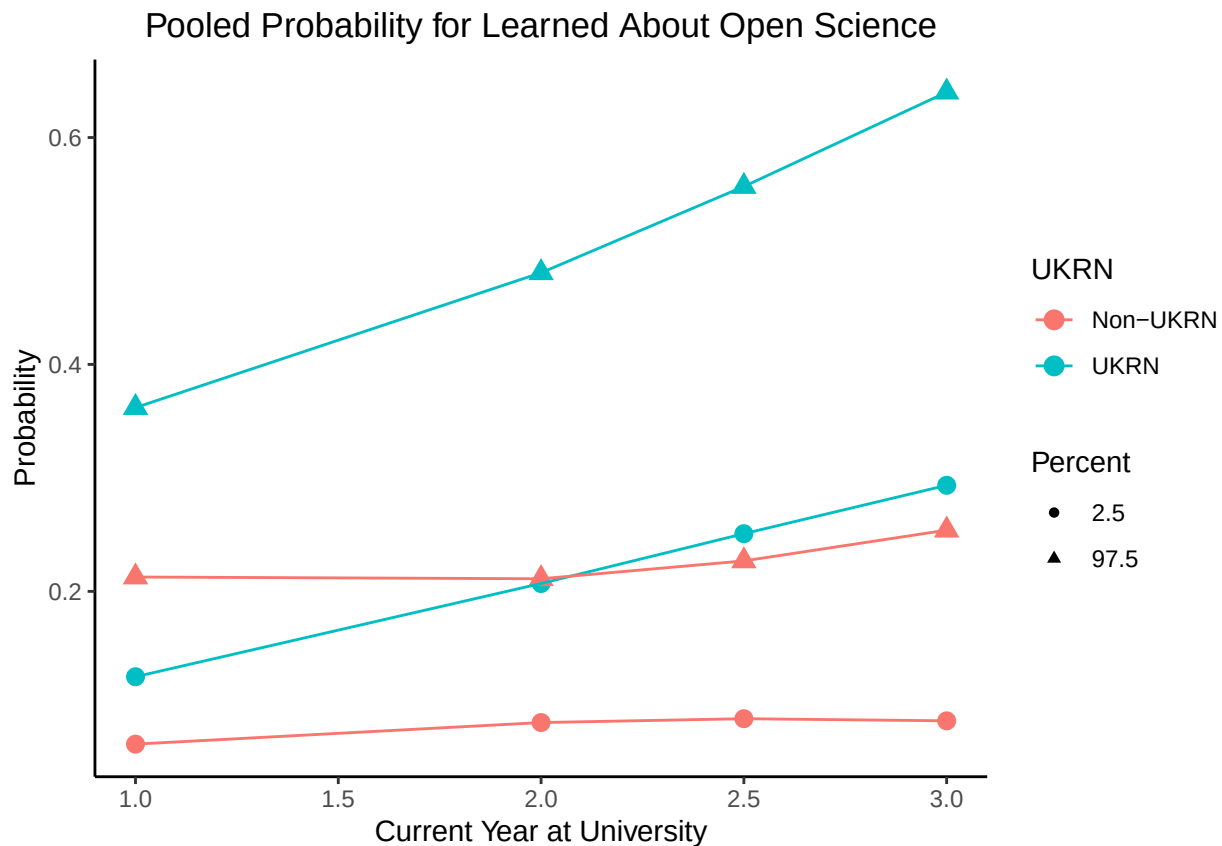
Year 1 N = 324

Year 2 N = 258

Placement Year N = 57

Year 4 N = 159

The output shows that all beta credibility intervals do cross 0. The results are non-significant.



Can explain open science

Bayesian Logistic regression for whether participants can explain what open science is (yes/no).

The explainer variables are:

- * Whether the participant was at a university that was affiliated with the UKRN
- * Whether the participant was in their first, second, placement or third year
- * Interaction between the two variables above

```
## Inference for Stan model: anon_model.
```

```
## 4 chains, each with iter=10000; warmup=5000; thin=1;
```

```
## post-warmup draws per chain=5000, total post-warmup draws=20000.
```

```
##
```

##	mean	se_mean	sd	2.5%	25%	50%	75%	97.5%	n_eff
## mu	-1.36	0.01	0.37	-2.08	-1.61	-1.36	-1.12	-0.63	1086
## sigma	0.37	0.01	0.14	0.12	0.27	0.36	0.46	0.70	774
## alpha[1]	-1.68	0.01	0.44	-2.56	-1.97	-1.68	-1.38	-0.83	1245
## alpha[2]	-1.47	0.01	0.48	-2.45	-1.78	-1.46	-1.15	-0.55	1561
## alpha[3]	-1.46	0.01	0.46	-2.39	-1.76	-1.45	-1.16	-0.57	1452
## alpha[4]	-1.16	0.01	0.47	-2.05	-1.48	-1.17	-0.86	-0.21	1531
## alpha[5]	-1.57	0.01	0.50	-2.63	-1.89	-1.55	-1.23	-0.63	1586
## alpha[6]	-1.08	0.01	0.49	-1.99	-1.41	-1.10	-0.77	-0.07	1641
## alpha[7]	-1.50	0.01	0.49	-2.51	-1.82	-1.49	-1.18	-0.53	1677
## alpha[8]	-0.96	0.01	0.46	-1.83	-1.28	-0.98	-0.67	-0.02	1429
## alpha[9]	-1.29	0.01	0.48	-2.23	-1.61	-1.29	-0.97	-0.33	1812
## alpha[10]	-1.37	0.01	0.48	-2.34	-1.68	-1.36	-1.05	-0.43	1805
## alpha[11]	-1.41	0.01	0.51	-2.44	-1.74	-1.40	-1.08	-0.40	1743
## alpha[12]	-1.64	0.01	0.53	-2.79	-1.96	-1.60	-1.28	-0.68	1839
## alpha[13]	-1.47	0.01	0.47	-2.44	-1.77	-1.46	-1.16	-0.57	1708
## alpha[14]	-1.48	0.01	0.48	-2.45	-1.78	-1.47	-1.16	-0.56	1749
## alpha[15]	-0.92	0.01	0.52	-1.87	-1.28	-0.95	-0.59	0.16	1728
## alpha[16]	-1.54	0.01	0.43	-2.42	-1.82	-1.53	-1.25	-0.71	1594
## alpha[17]	-1.43	0.01	0.45	-2.34	-1.73	-1.43	-1.14	-0.55	1736
## alpha[18]	-1.43	0.01	0.43	-2.29	-1.71	-1.42	-1.13	-0.59	1708
## alpha[19]	-1.58	0.01	0.50	-2.63	-1.90	-1.56	-1.24	-0.63	1756
## alpha[20]	-1.04	0.01	0.41	-1.83	-1.32	-1.05	-0.77	-0.23	1417
## alpha[21]	-1.17	0.01	0.41	-1.96	-1.45	-1.18	-0.90	-0.34	1612
## alpha[22]	-1.33	0.01	0.41	-2.13	-1.61	-1.33	-1.05	-0.51	1510
## alpha[23]	-1.35	0.01	0.50	-2.34	-1.68	-1.35	-1.03	-0.35	1976
## beta1	-0.03	0.01	0.50	-1.02	-0.37	-0.03	0.30	0.95	1488
## beta2	0.07	0.00	0.16	-0.25	-0.04	0.07	0.18	0.38	1187
## beta3	0.27	0.01	0.22	-0.16	0.12	0.27	0.42	0.71	1779
## lp__	-438.60	0.41	7.89	-452.51	-443.67	-439.18	-434.37	-419.79	379
##	Rhat								
## mu	1.00								
## sigma	1.01								
## alpha[1]	1.00								
## alpha[2]	1.00								
## alpha[3]	1.00								
## alpha[4]	1.00								
## alpha[5]	1.00								
## alpha[6]	1.00								
## alpha[7]	1.00								
## alpha[8]	1.00								
## alpha[9]	1.00								
## alpha[10]	1.00								
## alpha[11]	1.00								
## alpha[12]	1.00								
## alpha[13]	1.00								
## alpha[14]	1.00								
## alpha[15]	1.00								
## alpha[16]	1.00								
## alpha[17]	1.00								
## alpha[18]	1.00								
## alpha[19]	1.00								
## alpha[20]	1.00								
## alpha[21]	1.00								

```
## alpha[22] 1.00
## alpha[23] 1.00
## beta1     1.00
## beta2     1.00
## beta3     1.00
## lp__      1.01
##
## Samples were drawn using NUTS(diag_e) at Mon Sep 23 02:02:54 2024.
## For each parameter, n_eff is a crude measure of effective sample size,
## and Rhat is the potential scale reduction factor on split chains (at
## convergence, Rhat=1).
```

Number of participants in the analysis

Total N = 798

UKRN N = 420

Non-UKRN N = 378

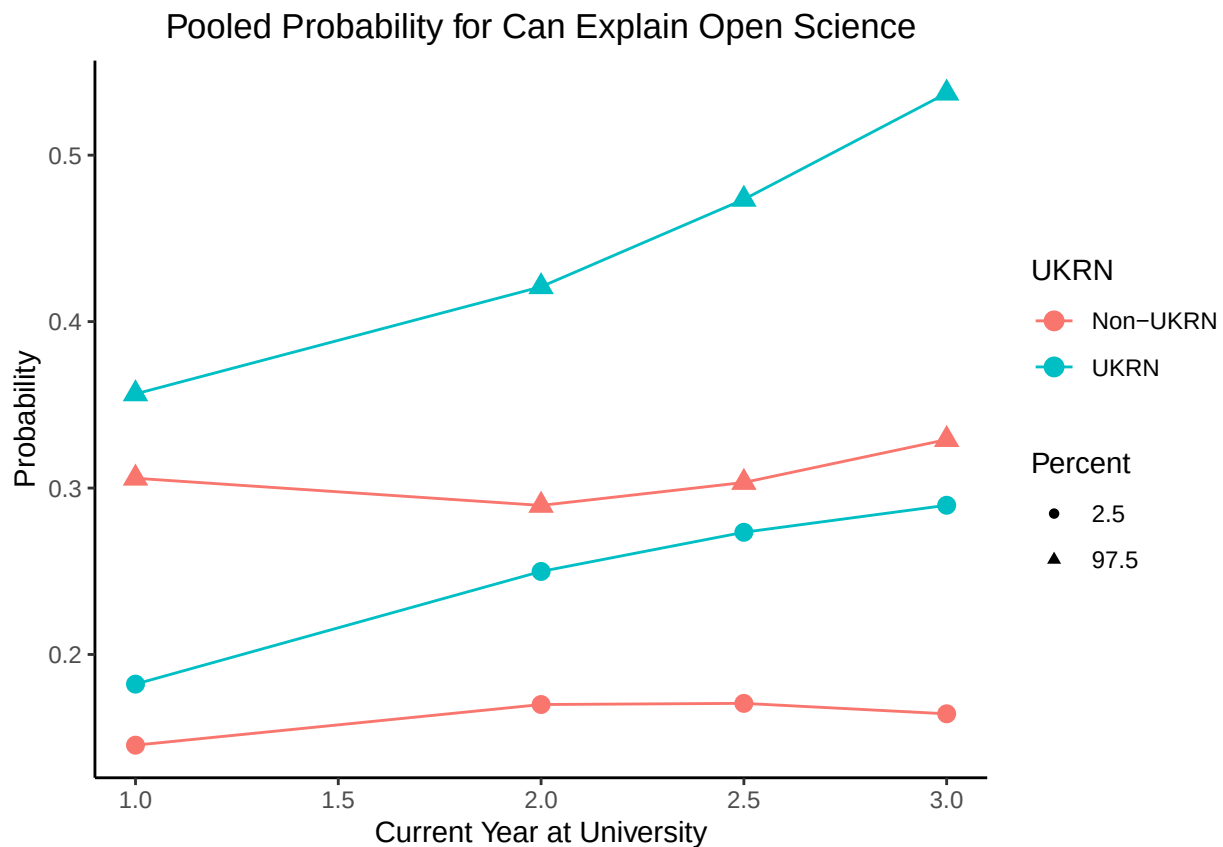
Year 1 N = 324

Year 2 N = 258

Placement Year N = 57

Year 4 N = 159

The output shows that all beta credibility intervals do cross 0. No significant results.



Applicability

Bayesian Logistic regression for whether participants think that the issues discussed in the questionnaire are relevant to them (yes/no). The explainer variables are:

* Whether the participant was at a university that was affiliated with the UKRN
 * Whether the participant was in their first, second, placement of third year
 * Interaction between the two variables above

```
## Inference for Stan model: anon_model.
## 4 chains, each with iter=10000; warmup=5000; thin=1;
## post-warmup draws per chain=5000, total post-warmup draws=20000.
##
##               mean se_mean   sd    2.5%    25%    50%    75%   97.5% n_eff
## mu             3.33    0.02 0.66    2.07    2.86    3.31    3.76    4.64  1660
## sigma          0.86    0.01 0.27    0.38    0.67    0.84    1.02    1.46  2659
## alpha[1]       3.02    0.02 0.78    1.51    2.49    3.01    3.54    4.59  1993
## alpha[2]       2.62    0.02 0.86    0.92    2.05    2.62    3.18    4.31  2397
## alpha[3]       3.50    0.02 0.87    1.84    2.91    3.48    4.08    5.27  2314
## alpha[4]       3.77    0.02 0.94    2.01    3.12    3.73    4.38    5.71  2625
## alpha[5]       2.55    0.02 0.87    0.82    1.98    2.56    3.14    4.25  2402
## alpha[6]       4.01    0.02 1.03    2.13    3.29    3.95    4.66    6.19  2816
## alpha[7]       3.41    0.02 0.95    1.61    2.77    3.38    4.01    5.37  2860
## alpha[8]       3.71    0.02 0.89    2.02    3.10    3.68    4.29    5.56  2508
## alpha[9]       3.43    0.02 0.90    1.75    2.81    3.40    4.01    5.30  2910
## alpha[10]      3.40    0.02 0.91    1.71    2.78    3.36    3.99    5.27  3016
## alpha[11]      3.34    0.02 0.95    1.54    2.70    3.32    3.97    5.26  3030
## alpha[12]      3.77    0.02 1.00    1.98    3.07    3.71    4.39    5.92  3345
## alpha[13]      2.05    0.02 0.77    0.59    1.52    2.03    2.56    3.64  2440
## alpha[14]      3.54    0.02 0.90    1.89    2.93    3.50    4.12    5.43  2960
## alpha[15]      3.36    0.02 0.91    1.67    2.73    3.33    3.95    5.26  3072
## alpha[16]      3.94    0.02 0.85    2.41    3.34    3.90    4.48    5.76  2815
## alpha[17]      3.54    0.02 0.85    1.98    2.96    3.50    4.08    5.34  3090
## alpha[18]      3.09    0.01 0.74    1.71    2.58    3.07    3.58    4.62  2814
## alpha[19]      2.64    0.02 0.83    1.05    2.08    2.62    3.19    4.27  2542
## alpha[20]      4.24    0.02 0.87    2.67    3.63    4.20    4.81    6.03  2575
## alpha[21]      4.27    0.02 0.96    2.59    3.60    4.20    4.86    6.40  2821
## alpha[22]      2.57    0.01 0.68    1.30    2.10    2.55    3.02    3.95  2183
## alpha[23]      2.75    0.02 0.92    1.01    2.13    2.74    3.36    4.60  3168
## beta1          -1.68    0.02 0.88   -3.37   -2.27   -1.67   -1.09    0.04  2270
## beta2          -0.37    0.01 0.26   -0.88   -0.54   -0.36   -0.19    0.13  1803
## beta3           0.88    0.01 0.38    0.15    0.63    0.88    1.14    1.65  3066
## lp__          -240.61    0.14 6.06 -252.53 -244.44 -240.64 -236.89 -228.64  1830
##
##               Rhat
## mu                1
## sigma             1
## alpha[1]          1
## alpha[2]          1
## alpha[3]          1
## alpha[4]          1
## alpha[5]          1
## alpha[6]          1
## alpha[7]          1
## alpha[8]          1
## alpha[9]          1
## alpha[10]         1
## alpha[11]         1
## alpha[12]         1
## alpha[13]         1
```

```

## alpha[14]      1
## alpha[15]      1
## alpha[16]      1
## alpha[17]      1
## alpha[18]      1
## alpha[19]      1
## alpha[20]      1
## alpha[21]      1
## alpha[22]      1
## alpha[23]      1
## beta1          1
## beta2          1
## beta3          1
## lp__           1
##
## Samples were drawn using NUTS(diag_e) at Mon Sep 23 02:05:20 2024.
## For each parameter, n_eff is a crude measure of effective sample size,
## and Rhat is the potential scale reduction factor on split chains (at
## convergence, Rhat=1).

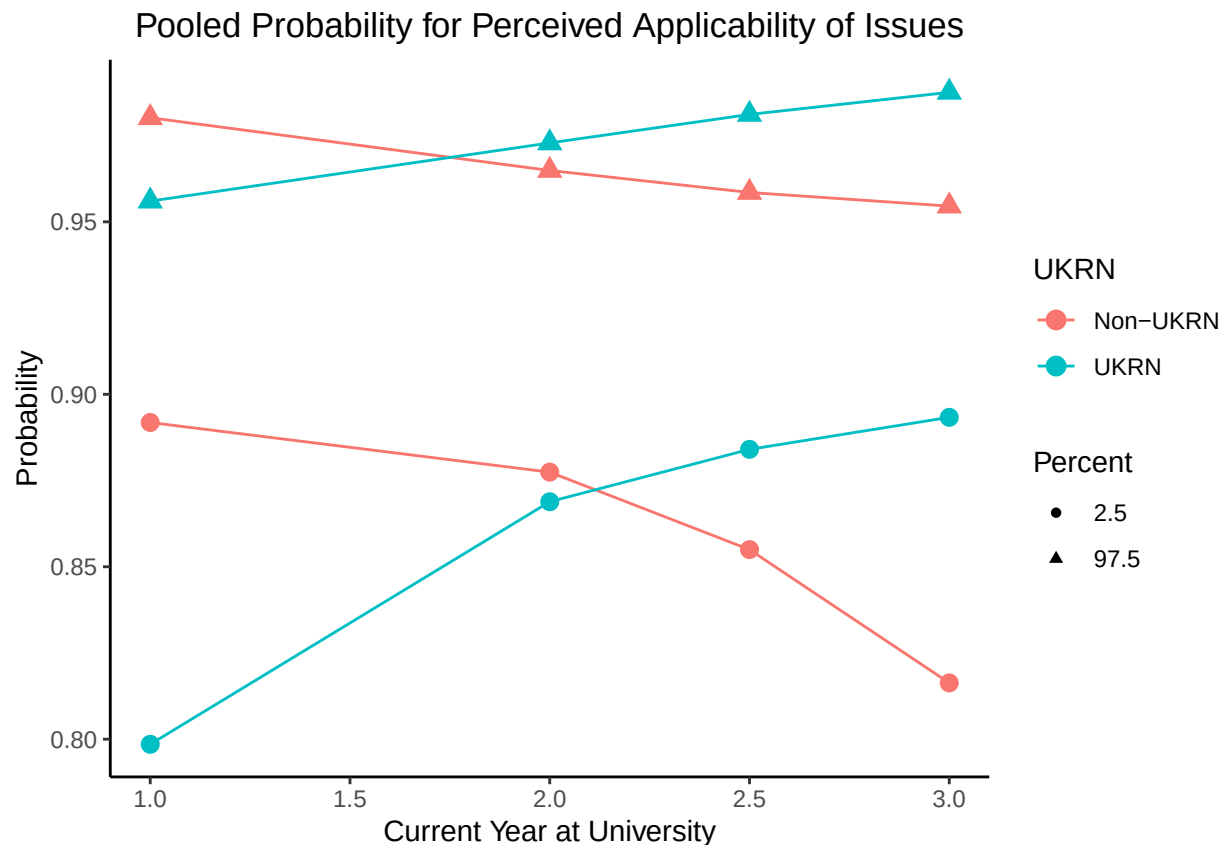
```

Number of participants in the analysis

Total N = 798
UKRN N = 420
Non-UKRN N = 378
Year 1 N = 324
Year 2 N = 258
Placement Year N = 57
Year 4 N = 159

The output shows that beta credibility intervals do not cross 0 only for the interaction term.
 2.5% = 0.1506118
 97.5% = 1.6540848

The interpretation is that students who attend a UKRN affiliated university are more likely to think that the issues around open and reproducible research are applicable to their work in the later years of education. For students who attend a non-UKRN affiliated university, there is no difference.



Data Set 2 - Truncated Poisson Regression

Awarweness Score Total

Bayesian regression for how much awareness of open and reproducible research the students had - scores range from 0 to 8.

The explainer variables are:

- * Whether the participant was at a university that was affiliated with the UKRN
- * Whether the participant was in their first, second, placement or third year
- * Interaction between the two variables above

```
## Inference for Stan model: anon_model.
## 4 chains, each with iter=10000; warmup=5000; thin=1;
## post-warmup draws per chain=5000, total post-warmup draws=20000.
##
##               mean se_mean   sd    2.5%    25%    50%    75%   97.5% n_eff
## mu           1.54      0.0 0.08    1.37    1.48    1.54    1.59    1.70  1520
## sigma        0.12      0.0 0.04     0.05    0.09    0.11    0.14    0.20  1842
## alpha[1]     1.51      0.0 0.10    1.31    1.45    1.51    1.58    1.72  1907
## alpha[2]     1.54      0.0 0.12    1.31    1.46    1.54    1.62    1.77  2522
## alpha[3]     1.52      0.0 0.11    1.30    1.44    1.52    1.59    1.74  2319
## alpha[4]     1.62      0.0 0.12    1.39    1.54    1.62    1.70    1.87  2400
## alpha[5]     1.43      0.0 0.12    1.18    1.35    1.44    1.52    1.66  2464
## alpha[6]     1.61      0.0 0.12    1.37    1.53    1.61    1.69    1.86  2445
## alpha[7]     1.50      0.0 0.12    1.26    1.42    1.51    1.59    1.74  2793
## alpha[8]     1.57      0.0 0.11    1.35    1.49    1.57    1.64    1.79  2319
## alpha[9]     1.51      0.0 0.12    1.28    1.44    1.52    1.59    1.75  3069
```

```

## alpha[10]    1.57    0.0 0.12    1.34    1.49    1.57    1.65    1.81    2931
## alpha[11]    1.57    0.0 0.13    1.33    1.49    1.57    1.66    1.83    2884
## alpha[12]    1.58    0.0 0.12    1.35    1.50    1.58    1.66    1.84    2999
## alpha[13]    1.45    0.0 0.11    1.22    1.38    1.46    1.53    1.67    2778
## alpha[14]    1.43    0.0 0.11    1.20    1.35    1.43    1.51    1.64    2655
## alpha[15]    1.63    0.0 0.13    1.39    1.54    1.62    1.71    1.89    2714
## alpha[16]    1.41    0.0 0.10    1.21    1.34    1.41    1.47    1.61    2005
## alpha[17]    1.38    0.0 0.12    1.14    1.30    1.38    1.46    1.60    2540
## alpha[18]    1.66    0.0 0.10    1.46    1.59    1.66    1.73    1.87    2445
## alpha[19]    1.52    0.0 0.11    1.29    1.45    1.52    1.60    1.75    2585
## alpha[20]    1.62    0.0 0.10    1.43    1.55    1.62    1.68    1.81    2060
## alpha[21]    1.57    0.0 0.09    1.38    1.51    1.57    1.63    1.76    2375
## alpha[22]    1.63    0.0 0.10    1.44    1.56    1.63    1.69    1.82    2131
## alpha[23]    1.52    0.0 0.12    1.28    1.43    1.52    1.60    1.75    3029
## beta1        -0.08    0.0 0.12   -0.31   -0.16   -0.08    0.00    0.16    2369
## beta2        -0.04    0.0 0.04   -0.11   -0.06   -0.04   -0.01    0.03    1572
## beta3         0.13    0.0 0.05    0.03    0.09    0.13    0.16    0.23    2741
## lp__         1892.40    0.2 6.05 1881.21 1888.53 1892.18 1895.83 1905.43    955
##              Rhat
## mu            1
## sigma          1
## alpha[1]       1
## alpha[2]       1
## alpha[3]       1
## alpha[4]       1
## alpha[5]       1
## alpha[6]       1
## alpha[7]       1
## alpha[8]       1
## alpha[9]       1
## alpha[10]      1
## alpha[11]      1
## alpha[12]      1
## alpha[13]      1
## alpha[14]      1
## alpha[15]      1
## alpha[16]      1
## alpha[17]      1
## alpha[18]      1
## alpha[19]      1
## alpha[20]      1
## alpha[21]      1
## alpha[22]      1
## alpha[23]      1
## beta1          1
## beta2          1
## beta3          1
## lp__           1
##
## Samples were drawn using NUTS(diag_e) at Mon Sep 23 02:18:15 2024.
## For each parameter, n_eff is a crude measure of effective sample size,
## and Rhat is the potential scale reduction factor on split chains (at
## convergence, Rhat=1).

```

Number of participants in the analysis

Total N = 842

UKRN N = 435

Non-UKRN N = 407

Year 1 N = 344

Year 2 N = 273

Placement Year N = 59

Year 4 N = 166

Var1	Freq
0	34
1	53
2	81
3	111
4	150
5	146
6	111
7	77
8	79

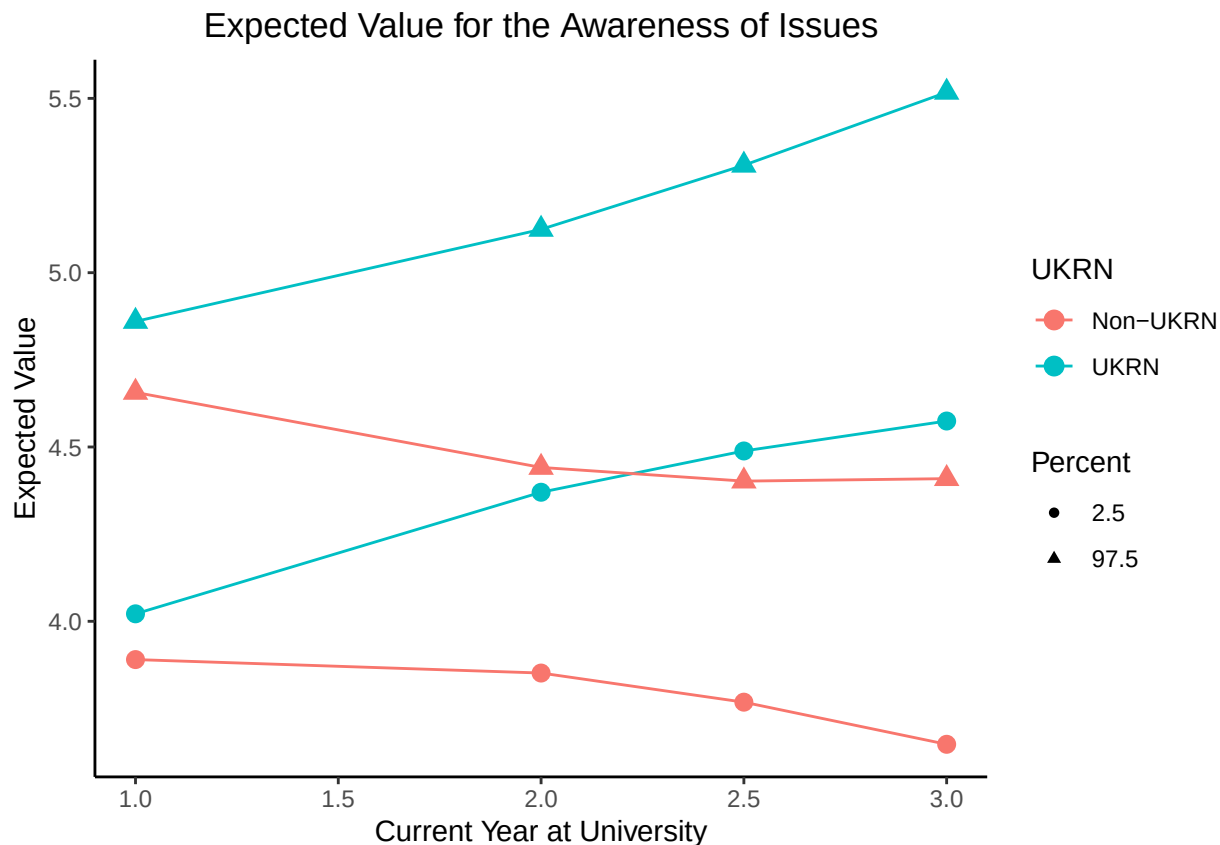
The output shows that beta credibility intervals do not cross 0 only for the interaction term.

2.5% = 0.0255122

97.5% = 0.2272617

The interpretation is that students who attend a UKRN affiliated university are more likely to be aware of the issues around open and reproducible research in the later years of education. For students who attend a non-UKRN affiliated university, there is no difference.

13 divergent transitions.



Experience Score Total

Bayesian regression for how much experience of open and reproducible research the students had - scores range from 0 to 8.

The explainer variables are:

- * Whether the participant was at a university that was affiliated with the UKRN
- * Whether the participant was in their first, second, placement or third year
- * Interaction between the two variables above

```
## Inference for Stan model: anon_model.
## 4 chains, each with iter=10000; warmup=5000; thin=1;
## post-warmup draws per chain=5000, total post-warmup draws=20000.
##
##               mean se_mean   sd    2.5%    25%    50%    75%   97.5%  n_eff
## mu           0.76    0.00  0.09    0.56    0.70    0.76    0.82    0.93   408
## sigma        0.06    0.00  0.04    0.01    0.03    0.06    0.09    0.15   276
## alpha[1]     0.78    0.00  0.11    0.56    0.71    0.78    0.85    0.98   468
## alpha[2]     0.74    0.00  0.11    0.50    0.68    0.75    0.82    0.95   564
## alpha[3]     0.76    0.00  0.11    0.53    0.69    0.76    0.83    0.96   510
## alpha[4]     0.78    0.00  0.11    0.55    0.71    0.78    0.85    0.99   545
## alpha[5]     0.72    0.01  0.12    0.47    0.65    0.73    0.80    0.93   544
## alpha[6]     0.76    0.00  0.11    0.52    0.69    0.76    0.83    0.98   549
## alpha[7]     0.71    0.01  0.12    0.44    0.64    0.72    0.79    0.92   537
## alpha[8]     0.82    0.01  0.12    0.60    0.74    0.81    0.90    1.06   469
## alpha[9]     0.72    0.01  0.12    0.47    0.65    0.73    0.80    0.93   533
## alpha[10]    0.71    0.01  0.12    0.44    0.64    0.72    0.79    0.93   559
```

```

## alpha[11]    0.74    0.00  0.12    0.50    0.67    0.75    0.82    0.96    602
## alpha[12]    0.78    0.00  0.11    0.55    0.70    0.78    0.85    1.00    591
## alpha[13]    0.76    0.00  0.11    0.53    0.69    0.76    0.83    0.96    563
## alpha[14]    0.75    0.00  0.11    0.52    0.68    0.75    0.82    0.95    564
## alpha[15]    0.77    0.00  0.11    0.53    0.70    0.77    0.84    0.98    583
## alpha[16]    0.79    0.00  0.10    0.58    0.72    0.79    0.86    0.99    506
## alpha[17]    0.77    0.00  0.11    0.55    0.70    0.77    0.84    0.97    532
## alpha[18]    0.76    0.00  0.10    0.55    0.69    0.76    0.82    0.95    530
## alpha[19]    0.76    0.00  0.11    0.52    0.69    0.76    0.83    0.97    548
## alpha[20]    0.76    0.00  0.10    0.55    0.69    0.76    0.83    0.95    491
## alpha[21]    0.77    0.00  0.10    0.57    0.71    0.77    0.84    0.97    522
## alpha[22]    0.74    0.00  0.10    0.51    0.67    0.74    0.80    0.93    487
## alpha[23]    0.81    0.01  0.12    0.58    0.73    0.80    0.89    1.07    560
## beta1        -0.13    0.01  0.13   -0.39   -0.22   -0.13   -0.04    0.13    451
## beta2         0.06    0.00  0.04   -0.02    0.03    0.06    0.08    0.15    474
## beta3         0.06    0.00  0.06   -0.05    0.02    0.06    0.11    0.18    516
## lp__         -211.03    1.08 13.56 -232.49 -221.27 -213.18 -201.50 -183.27    157
##              Rhat
## mu           1.00
## sigma        1.02
## alpha[1]     1.00
## alpha[2]     1.00
## alpha[3]     1.00
## alpha[4]     1.00
## alpha[5]     1.00
## alpha[6]     1.00
## alpha[7]     1.00
## alpha[8]     1.00
## alpha[9]     1.00
## alpha[10]    1.00
## alpha[11]    1.00
## alpha[12]    1.00
## alpha[13]    1.00
## alpha[14]    1.00
## alpha[15]    1.00
## alpha[16]    1.00
## alpha[17]    1.00
## alpha[18]    1.00
## alpha[19]    1.00
## alpha[20]    1.00
## alpha[21]    1.00
## alpha[22]    1.00
## alpha[23]    1.00
## beta1        1.00
## beta2        1.00
## beta3        1.00
## lp__         1.03
##
## Samples were drawn using NUTS(diag_e) at Mon Sep 23 02:32:47 2024.
## For each parameter, n_eff is a crude measure of effective sample size,
## and Rhat is the potential scale reduction factor on split chains (at
## convergence, Rhat=1).

```

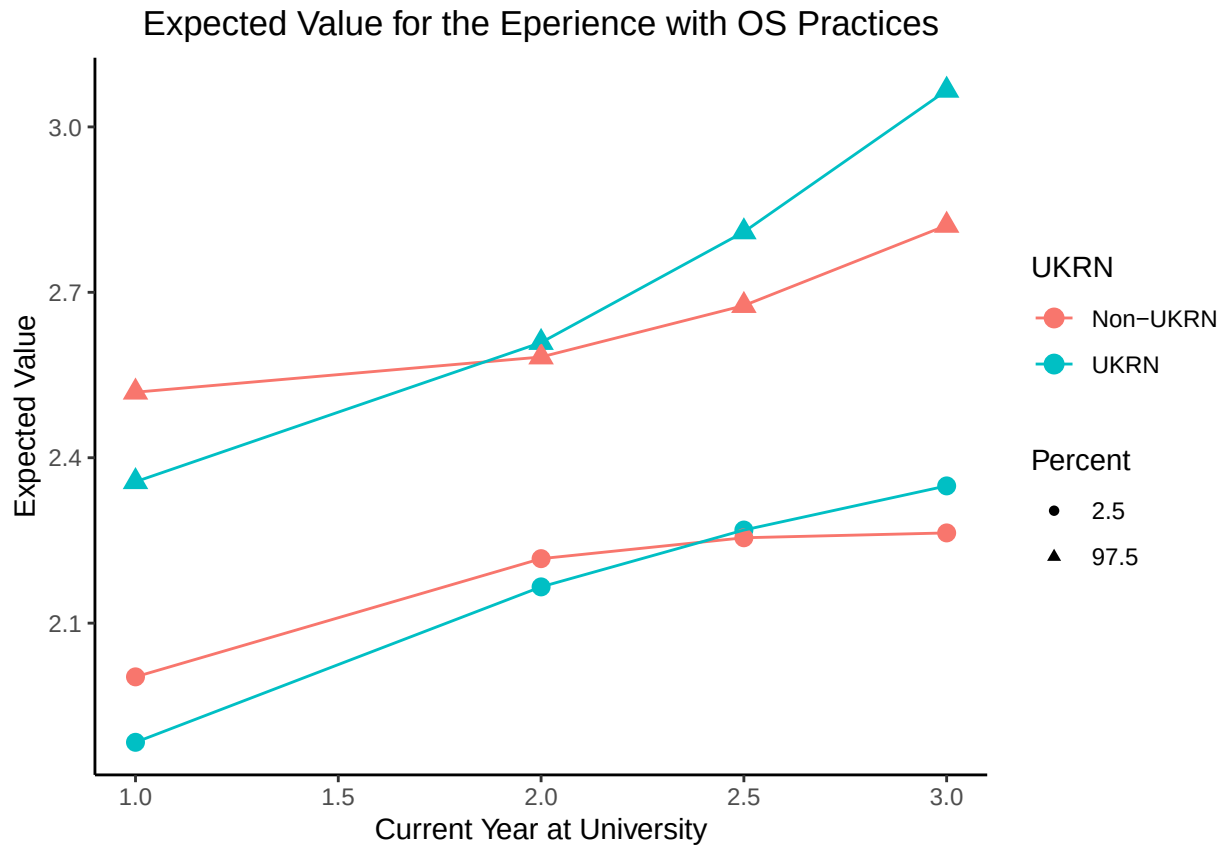
Number of participants in the analysis

Total N = 842
 UKRN N = 435
 Non-UKRN N = 407
 Year 1 N = 344
 Year 2 N = 273
 Placement Year N = 59
 Year 4 N = 166

Var1	Freq
0	138
1	169
2	159
3	165
4	103
5	65
6	28
7	9
8	6

The output shows that all beta credibility intervals cross 0.

2400 divergent transitions. In addition: - There were 2 chains where the estimated Bayesian Fraction of Missing Information was low. - The largest R-hat is 1.05, indicating chains have not mixed. Running the chains for more iterations may help. - Effective Samples Size (ESS) is too low, indicating posterior means and medians may be unreliable. - Effective Samples Size (ESS) is too low, indicating posterior variances and tail quantiles may be unreliable.



Data Set 2 - Linear Regression

Conceptual Perception Score

Bayesian regression for how positive are students perceptions of open and reproducible research. The scores are between -2 and 2 but will be changed by adding 2.001. This way the scale will be from 0 to 4.

The explainer variables are:

* Whether the participant was at a university that was affiliated with the UKRN

* Whether the participant was in their first, second, placement or third year

* Interaction between the two variables above

```
## Inference for Stan model: anon_model.
## 4 chains, each with iter=10000; warmup=5000; thin=1;
## post-warmup draws per chain=5000, total post-warmup draws=20000.
##
##               mean se_mean   sd  2.5%  25%   50%   75%  97.5% n_eff Rhat
## mu           0.58    0.00 0.05   0.47  0.54  0.58   0.61   0.68 1294 1.00
## sigma        0.35    0.00 0.01   0.33  0.34  0.35   0.35   0.36 1971 1.00
## sigma_a      0.07    0.00 0.02   0.02  0.05  0.07   0.08   0.12 1232 1.00
## alpha[1]     0.52    0.00 0.07   0.39  0.48  0.52   0.57   0.65 1473 1.00
## alpha[2]     0.53    0.00 0.08   0.37  0.48  0.53   0.58   0.67 1984 1.00
## alpha[3]     0.58    0.00 0.07   0.44  0.53  0.58   0.62   0.71 1888 1.00
## alpha[4]     0.60    0.00 0.07   0.45  0.55  0.60   0.64   0.74 2118 1.00
## alpha[5]     0.50    0.00 0.08   0.34  0.45  0.51   0.56   0.65 1684 1.00
## alpha[6]     0.68    0.00 0.09   0.53  0.62  0.68   0.74   0.86 2265 1.00
## alpha[7]     0.58    0.00 0.08   0.43  0.53  0.58   0.63   0.73 2131 1.00
## alpha[8]     0.61    0.00 0.07   0.48  0.57  0.61   0.66   0.75 1983 1.00
## alpha[9]     0.55    0.00 0.07   0.41  0.50  0.56   0.60   0.70 2241 1.00
## alpha[10]    0.56    0.00 0.07   0.41  0.51  0.56   0.61   0.70 2085 1.00
## alpha[11]    0.59    0.00 0.08   0.44  0.54  0.59   0.64   0.75 2296 1.00
## alpha[12]    0.52    0.00 0.08   0.35  0.47  0.52   0.57   0.67 2026 1.00
## alpha[13]    0.60    0.00 0.07   0.46  0.55  0.60   0.64   0.74 2295 1.00
## alpha[14]    0.53    0.00 0.07   0.39  0.49  0.54   0.58   0.67 1758 1.00
## alpha[15]    0.60    0.00 0.08   0.45  0.55  0.60   0.65   0.75 2526 1.00
## alpha[16]    0.59    0.00 0.06   0.47  0.55  0.59   0.64   0.72 2070 1.00
## alpha[17]    0.55    0.00 0.07   0.42  0.51  0.56   0.60   0.68 2251 1.00
## alpha[18]    0.55    0.00 0.06   0.43  0.51  0.55   0.59   0.67 1943 1.00
## alpha[19]    0.58    0.00 0.07   0.43  0.53  0.58   0.62   0.71 2131 1.00
## alpha[20]    0.63    0.00 0.06   0.51  0.59  0.63   0.67   0.75 1747 1.00
## alpha[21]    0.60    0.00 0.06   0.48  0.56  0.60   0.64   0.72 2139 1.00
## alpha[22]    0.62    0.00 0.06   0.50  0.58  0.62   0.66   0.74 1862 1.00
## alpha[23]    0.55    0.00 0.08   0.39  0.50  0.55   0.60   0.69 2305 1.00
## beta1       -0.04    0.00 0.08  -0.19 -0.09 -0.04   0.02   0.12 1757 1.00
## beta2        0.01    0.00 0.02  -0.04 -0.01  0.01   0.02   0.05 1356 1.00
## beta3        0.05    0.00 0.03  -0.01  0.03  0.05   0.08   0.12 2024 1.00
## lp__        518.27    0.34 7.19 505.99 513.77 517.69 521.84 535.59 435 1.01
##
## Samples were drawn using NUTS(diag_e) at Mon Sep 23 02:46:50 2024.
## For each parameter, n_eff is a crude measure of effective sample size,
## and Rhat is the potential scale reduction factor on split chains (at
## convergence, Rhat=1).
```

Number of participants in the analysis

Total N = 835

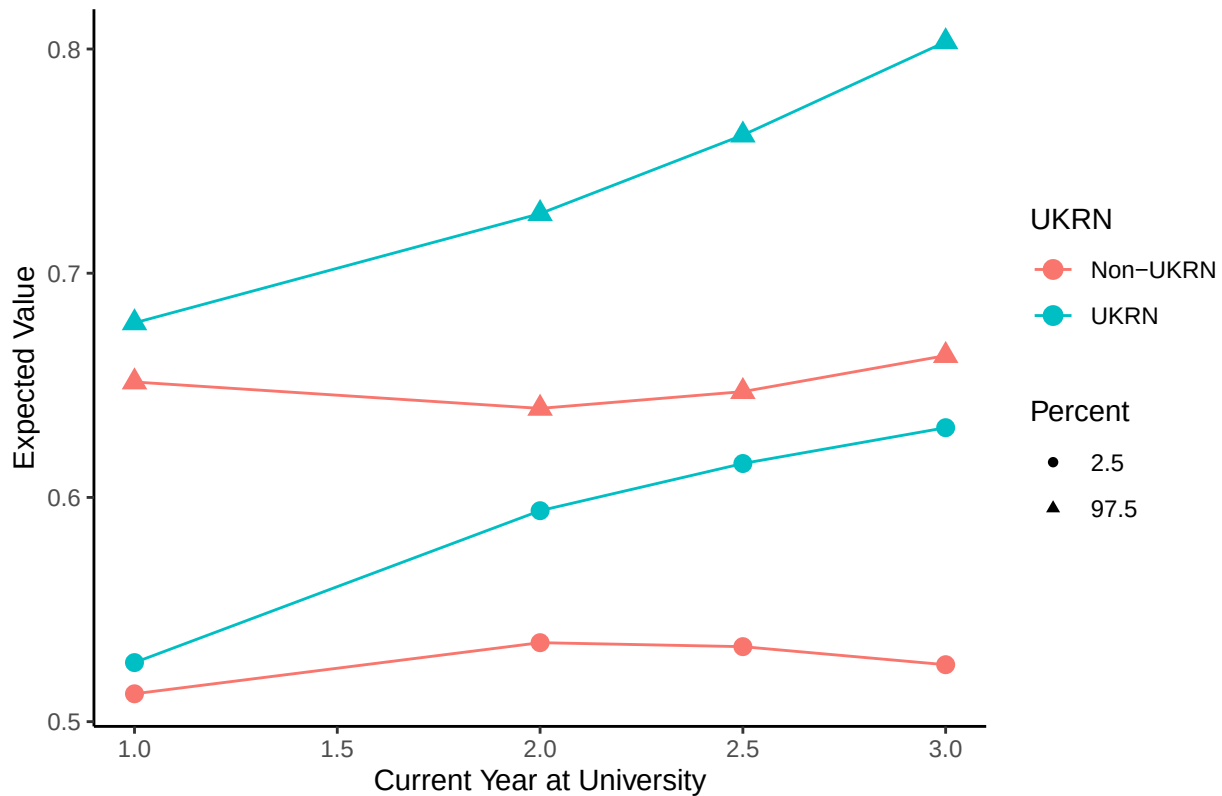
UKRN N = 432

Non-UKRN N = 403
 Year 1 N = 340
 Year 2 N = 270
 Placement Year N = 59
 Year 4 N = 166

The output shows that all beta credibility intervals cross 0.

46 divergent transitions.

Expected Value for positive Conceptual Perceptions of OS Practices



Situational Perception Score

Bayesian regression for how positive are students perceptions of open and reproducible research. The scores are between -2 and 2 but will be changed by adding 2.001. This way the scale will be from 0 to 4.

The explainer variables are:

- * Whether the participant was at a university that was affiliated with the UKRN
- * Whether the participant was in their first, second, placement or third year
- * Interaction between the two variables above

```
## Inference for Stan model: anon_model.
## 4 chains, each with iter=10000; warmup=5000; thin=1;
## post-warmup draws per chain=5000, total post-warmup draws=20000.
##
##          mean se_mean  sd  2.5%  25%  50%  75%  97.5% n_eff Rhat
## mu          0.36    0.00 0.08   0.20  0.31  0.36  0.41   0.52 2406   1
## sigma       0.48    0.00 0.01   0.46  0.47  0.48  0.49   0.51 17329  1
## sigma_a     0.14    0.00 0.04   0.08  0.12  0.14  0.16   0.23  7101  1
## alpha[1]    0.20    0.00 0.11  -0.02  0.13  0.20  0.27   0.41  2911  1
```

```

## alpha[2]    0.24    0.00 0.13 -0.01    0.16    0.24    0.33    0.48 3811    1
## alpha[3]    0.30    0.00 0.12  0.08    0.23    0.30    0.38    0.53 3413    1
## alpha[4]    0.40    0.00 0.12  0.17    0.32    0.40    0.48    0.64 3700    1
## alpha[5]    0.19    0.00 0.13 -0.08    0.11    0.19    0.28    0.44 3988    1
## alpha[6]    0.61    0.00 0.13  0.36    0.52    0.61    0.69    0.88 4285    1
## alpha[7]    0.49    0.00 0.13  0.24    0.40    0.49    0.57    0.76 4390    1
## alpha[8]    0.44    0.00 0.12  0.22    0.37    0.44    0.52    0.68 3438    1
## alpha[9]    0.31    0.00 0.12  0.08    0.23    0.31    0.40    0.55 4734    1
## alpha[10]   0.40    0.00 0.12  0.16    0.32    0.40    0.48    0.65 5088    1
## alpha[11]   0.42    0.00 0.13  0.17    0.34    0.42    0.51    0.69 5145    1
## alpha[12]   0.25    0.00 0.13 -0.01    0.16    0.25    0.34    0.50 5313    1
## alpha[13]   0.40    0.00 0.11  0.18    0.33    0.40    0.47    0.61 4949    1
## alpha[14]   0.30    0.00 0.11  0.08    0.23    0.31    0.38    0.52 4846    1
## alpha[15]   0.44    0.00 0.12  0.20    0.35    0.43    0.52    0.69 5340    1
## alpha[16]   0.36    0.00 0.09  0.18    0.30    0.36    0.43    0.55 3969    1
## alpha[17]   0.24    0.00 0.11  0.03    0.17    0.24    0.32    0.46 5104    1
## alpha[18]   0.37    0.00 0.09  0.19    0.30    0.37    0.43    0.56 4855    1
## alpha[19]   0.28    0.00 0.12  0.05    0.20    0.28    0.36    0.51 4457    1
## alpha[20]   0.44    0.00 0.09  0.26    0.38    0.44    0.50    0.61 3576    1
## alpha[21]   0.39    0.00 0.09  0.21    0.33    0.39    0.45    0.57 4422    1
## alpha[22]   0.51    0.00 0.09  0.33    0.45    0.51    0.57    0.69 3787    1
## alpha[23]   0.28    0.00 0.13  0.02    0.20    0.29    0.37    0.53 5526    1
## beta1       -0.02    0.00 0.12 -0.26   -0.10   -0.03    0.05    0.22 3277    1
## beta2       -0.01    0.00 0.03 -0.08   -0.04   -0.01    0.01    0.05 2591    1
## beta3        0.12    0.00 0.05  0.02    0.09    0.12    0.15    0.21 4170    1
## lp__        218.16    0.06 4.52 208.34 215.35 218.46 221.31 226.14 5513    1
##
## Samples were drawn using NUTS(diag_e) at Mon Sep 23 02:58:09 2024.
## For each parameter, n_eff is a crude measure of effective sample size,
## and Rhat is the potential scale reduction factor on split chains (at
## convergence, Rhat=1).

```

Number of participants in the analysis

Total N = 803

UKRN N = 421

Non-UKRN N = 382

Year 1 N = 326

Year 2 N = 261

Placement Year N = 57

Year 4 N = 159

The output shows that beta credibility intervals do not cross 0 only for the interaction term.

2.5% = 0.0241971

97.5% = 0.2115566

The interpretation is that students who attend a UKRN affiliated university are more likely to have positive situational perceptions of open and reproducible practices in the later years of education. For students who attend a non-UKRN affiliated university, there is no difference.

Expected Value for positive Conceptual Perceptions of OS Practices

